

Model Training Diagnostics

Business Report | Treatment Outcome Prediction | LSTM vs. Neural Network Comparison

Executive Summary

Two deep learning architectures — a sequential LSTM model and a standard feed-forward neural network (MLP) — were trained on the clinical dataset to predict treatment outcome. Both models were evaluated using training and validation loss and AUC (Area Under the ROC Curve) curves across training epochs. This report summarizes the training behavior, diagnoses underfitting/overfitting patterns, and provides recommendations for improving model performance.

Metric	LSTM Model	Neural Network (MLP)
Final Train AUC	~0.648	~0.646
Final Validation AUC	~0.647	~0.648
Final Train Loss	~0.659	~0.660
Final Validation Loss	~0.655 (unstable)	~0.664 (mildly noisy)
Train/Val Convergence	Weak — validation is noisy	Strong — curves converge closely
Diagnosis	Underfitting + instability	Underfitting, but stable (near sweet spot)

Bottom line: Neither model is overfitting. Both plateau at a modest AUC (~0.65), which is only moderately better than random guessing (0.50). This points to a ceiling driven by limited predictive signal in the current features/data, not by excessive model complexity. The Neural Network trains more stably; the LSTM adds architectural complexity without a performance gain, which is expected since the dataset is one row per patient (sequence length of 1) — there is no real temporal sequence for an LSTM to exploit.

1. LSTM Model — Training Diagnostics

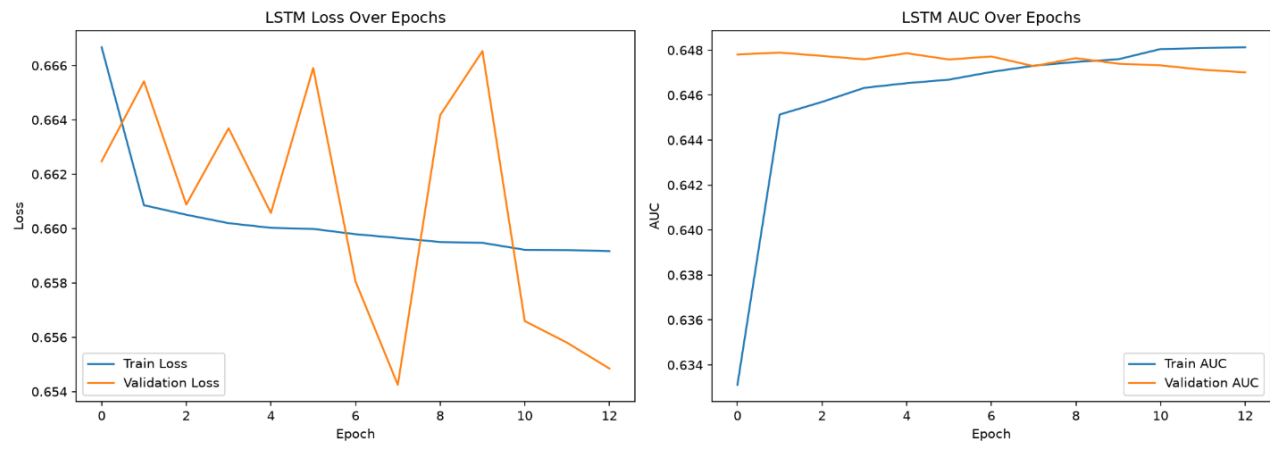


Figure 1. LSTM training and validation loss (left) and AUC (right) over 13 epochs.

Observations

- **Train loss** decreases smoothly from ~0.667 to a stable plateau around 0.659, indicating the model is optimizing normally on the training data.
- **Validation loss** is highly volatile, oscillating between ~0.654 and ~0.667 with no consistent downward trend, and never converges toward the training loss.
- **Train and validation AUC** track much more closely and plateau together around 0.646–0.648, showing the AUC metric is less sensitive to the same noise seen in loss.
- The gap between train and validation AUC stays small and does not widen over training — there is no classic overfitting signature (train improving while validation degrades).

Diagnosis: Underfitting with Validation Instability

Both metrics plateau at a modest performance level rather than diverging, which is the signature of **underfitting**, not overfitting. The choppy validation loss curve suggests additional volatility layered on top — likely caused by the small validation set size (20% of an already-limited patient-level dataset), the relatively high batch size (2048) relative to dataset size, or the use of a sequence length of 1, which removes the LSTM's core advantage (modeling temporal dependencies) and leaves it functioning like an over-parameterized dense network.

2. Neural Network (MLP) — Training Diagnostics

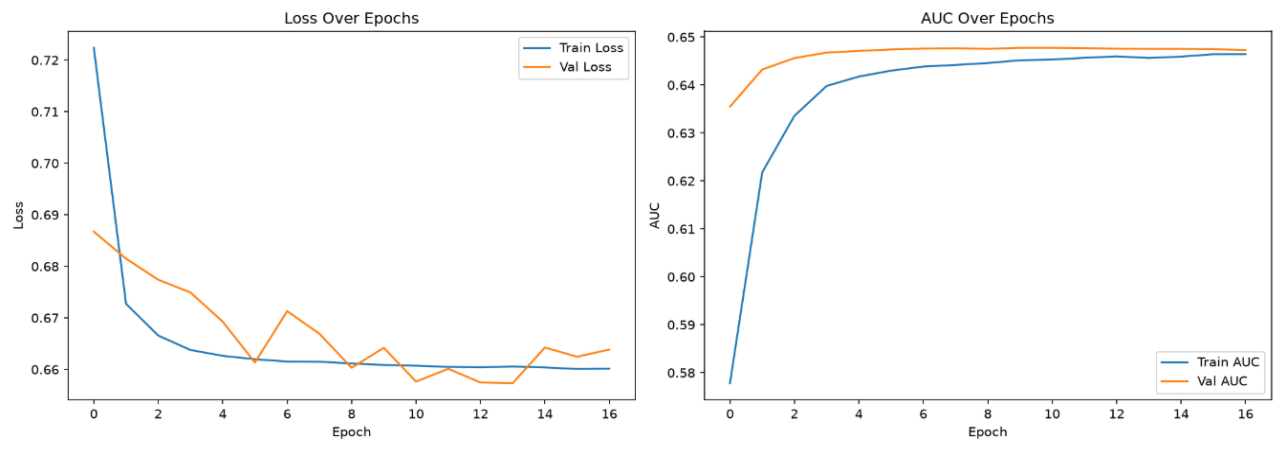


Figure 2. Neural network training and validation loss (left) and AUC (right) over 17 epochs.

Observations

- **Train loss** drops sharply in the first 2 epochs ($0.72 \rightarrow 0.66$) and then flattens, remaining stable through epoch 17.
- **Validation loss** follows a similar early drop and stays close to the training curve throughout, with only mild wobble (roughly 0.657–0.671) rather than the wide, erratic swings seen in the LSTM.
- **Train and validation AUC** converge tightly and plateau together around 0.645–0.648, with validation AUC slightly ahead of training AUC for most of training — a sign the validation split is representative rather than harder than the training data.
- No divergence or widening gap appears between train and validation curves at any point in training.

Diagnosis: Near Sweet-Spot Convergence, Underfit Ceiling

This model shows the healthiest training dynamic of the two: train and validation curves move together and plateau at essentially the same values, which is the textbook signature of a well-fit training process (the “sweet spot” in terms of the bias-variance tradeoff). However, the *absolute* AUC value it plateaus at (~ 0.65) is still low, meaning the model has converged to a genuine but limited amount of predictive signal in the data — this is a case of stable training combined with an underfitting-level performance ceiling.

3. Comparative Analysis

Aspect	LSTM	Neural Network (MLP)
Training stability	Noisy validation curve	Smooth, closely-tracked curves
Train/Val gap	Small on AUC, erratic on loss	Consistently small on both metrics
Peak Validation AUC	~0.648	~0.648
Architecture fit for data	Poor — sequence length = 1 negates LSTM's core strength	Appropriate — data is tabular, one row per patient
Recommended for production	Not preferred — added complexity, no AUC gain	Preferred — simpler, faster, equally accurate, more stable

A key structural note: the underlying LSTM script (10_LSTM_Sequential_Model.py) uses **SEQUENCE_LENGTH = 1**, meaning each patient contributes a single timestep rather than a true sequence of clinical events over time. An LSTM's main advantage — learning temporal/sequential dependencies — cannot be exploited under this configuration, which explains why it performs no better than the simpler feed-forward network despite being architecturally more complex (stacked LSTM layers, dropout, more parameters).

4. Business Implications

- **Current models are not yet reliable enough for clinical decision support.** An AUC of ~0.65 indicates limited discriminative power between patients who will and won't respond to treatment; deploying either model as-is risks a high rate of misclassified outcomes.
- **The bottleneck is data/feature-driven, not model-driven.** Since both a simple and a complex architecture converge to nearly identical performance, adding more model complexity (e.g., deeper networks, more LSTM layers) is unlikely to improve results without first improving the input features.
- **The simpler Neural Network is the more practical choice today.** It achieves equal AUC with substantially more stable training, lower computational cost, and no architectural mismatch with the data structure.

5. Recommendations

- **Improve feature engineering** — revisit clinical features for interaction terms, derived risk scores, or domain-specific transformations that could raise the ceiling on predictive signal.
- **Re-evaluate the case for using an LSTM at all** — unless true longitudinal (multi-visit) patient data becomes available with a genuine time dimension, a feed-forward network or gradient-boosted tree model is a better architectural fit.

- **Stabilize LSTM validation behavior** if it is retained — consider a larger validation set, a lower learning rate, or a smaller batch size to reduce the noisy validation loss swings observed in Figure 1.
- **Report test-set performance alongside training curves** — the current plots show train/validation only; final unbiased performance should be confirmed against the held-out test set (see 10_lstm_treatment_outcome_metrics.txt and the equivalent file for the neural network) before any deployment decision.
- **Benchmark against tree-based models** (e.g., XGBoost, Gradient Boosting, already present in this project's output) which often outperform neural networks on small-to-medium tabular clinical datasets and may close the AUC gap more effectively than further neural architecture changes.